12.432

EE25BTECH11002 - Achat Parth Kalpesh

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Question

If any two columns of $\mathbf{p} = \begin{pmatrix} 4 & 7 & 8 \\ 3 & 1 & 5 \\ 9 & 6 & 2 \end{pmatrix}$ are interchanged, which one of the following statements regarding the value of the determinant is CORRECT?

1. Absolute value remains unchanged but sign will change.
2. Both absolute value and sign will change.
3. Absolute value will change but sign will not change.
4. Both absolute value and sign will remain unchanged.

Solution

Let

$$\mathbf{P} = [\mathbf{p}_1 \ \mathbf{p}_2 \ \mathbf{p}_3] \tag{1}$$

where

$$\mathbf{p}_1 = \begin{pmatrix} 4 \\ 3 \\ 9 \end{pmatrix}, \quad \mathbf{p}_2 = \begin{pmatrix} 7 \\ 1 \\ 6 \end{pmatrix}, \quad \mathbf{p}_3 = \begin{pmatrix} 8 \\ 5 \\ 2 \end{pmatrix} \tag{2}$$

Then

$$\det(\mathbf{P}) = \det(\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3) \tag{3}$$

Solution

Determinant of $\mathbf{P}$ is

$$\det\left(\mathbf{P}\right) = 4\left(1\cdot 2 - 5\cdot 6\right) - 7\left(3\cdot 2 - 5\cdot 9\right) + 8\left(3\cdot 6 - 1\cdot 9\right) \tag{4}$$

$$= 4\left(2 - 30\right) - 7\left(6 - 45\right) + 8\left(18 - 9\right) \tag{5}$$

$$= 4\left(-28\right) - 7\left(-39\right) + 8\left(9\right) \tag{6}$$

$$= -112 + 273 + 72 \tag{7}$$

$$= 233 \tag{8}$$

Hence,

$$\det\left(\mathbf{P}\right) = 233 \tag{9}$$

Determinant after interchanging columns 1 and 2 is $\mathbf{P}'$

$$\mathbf{P}' = \begin{bmatrix}\mathbf{p}_2 & \mathbf{p}_1 & \mathbf{p}_3\end{bmatrix} = \begin{pmatrix} 7 & 4 & 8 \\ 1 & 3 & 5 \\ 6 & 9 & 2 \end{pmatrix} \tag{10}$$

Solution

Then

$$\det\left(\mathbf{P}'\right) = 7\left(3\cdot 2 - 5\cdot 9\right) - 4\left(1\cdot 2 - 5\cdot 6\right) + 8\left(1\cdot 9 - 3\cdot 6\right) \tag{11}$$
$$= 7\left(6 - 45\right) - 4\left(2 - 30\right) + 8\left(9 - 18\right) \tag{12}$$
$$= 7\left(-39\right) - 4\left(-28\right) + 8\left(-9\right) \tag{13}$$
$$= -273 + 112 - 72 \tag{14}$$
$$= -233 \tag{15}$$

Thus,

$$\det\left(\mathbf{P}'\right) = -233 \tag{16}$$

Solution

$$\det\left(\mathbf{P}'\right) = -\det\left(\mathbf{P}\right) \tag{17}$$

Hence, on interchanging any two columns of a matrix:
The sign of the determinant changes, but the absolute value remains the same